

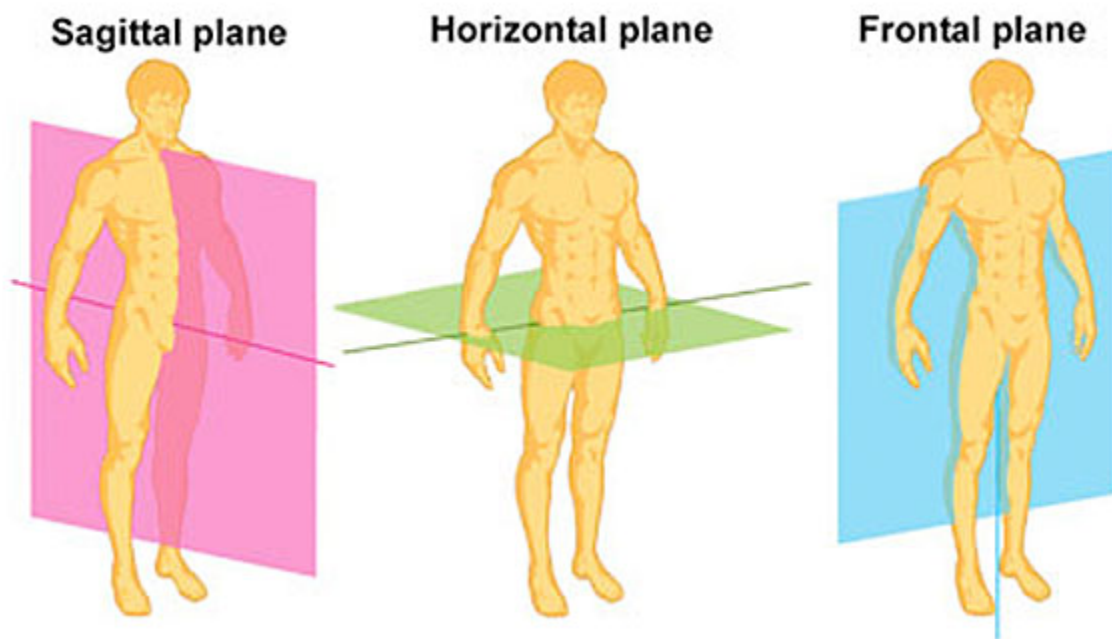
Shoulder Abduction

Performance:

Patient stands upright, back straight, feet shoulder-width apart Both arms resting at sides, elbows fully straight Palms facing inward toward the body (neutral grip) or forward (anatomical) Patient raises one arm outward and upward in a smooth arc — away from the body and to the side, not forward. Continues to a comfortable end range (90° horizontal or higher). Hold briefly at the top. Lowers back down slowly and with control Returns arm fully to side before next rep.

The movement is entirely in the FRONTAL PLANE — the arm travels directly out to the side, staying in line with the torso, not drifting forward or backward. This is the defining distinction between shoulder abduction and shoulder flexion.

Both look like raising an arm. The difference: Flexion → arm travels forward and upward (sagittal plane) Abduction → arm travels sideways and upward (frontal plane)



Camera:

View → strict FRONT view — camera faces the patient directly this is the opposite of all your previous exercises (heel slides, straight leg raise, shoulder flexion) which all used side view.

Camera height → shoulder height of patient (~130-150cm from floor) same reasoning as shoulder flexion — the arm travels from below shoulder height to above it; a centred camera captures the full arc symmetrically.

Camera distance → 2 to 2.5 metres from patient must be fully in frame including when arm is at full abduction — check this before recording.

Background → plain wall, minimal clutter Lighting → even, diffuse. Source in front of patient (not behind) Clothing → fitted short-sleeved or sleeveless top.

Patient position → facing the camera directly feet shoulder-width apart moving arm toward the camera side — either arm works since both are visible from front view the stationary arm should remain relaxed at the side.

Which side to film: Unlike your side-view exercises where you must film from the active-limb side, front-view captures both arms simultaneously. You still specify which arm is the active one in your code, but the camera position does not change between left and right.

Why front view specifically: Abduction is a frontal-plane movement. From the side, the arm appears to barely move — you are looking down the axis of the motion, not across it. The shoulder-to-elbow vector has almost no horizontal displacement from the side view as the arm abducts. You cannot measure the abduction angle from the side any more than you could measure knee flexion from the front. From the front, the arm sweeps clearly from vertical (at-side) to horizontal (90°) to overhead, and the angle between the trunk and the arm is directly measurable.

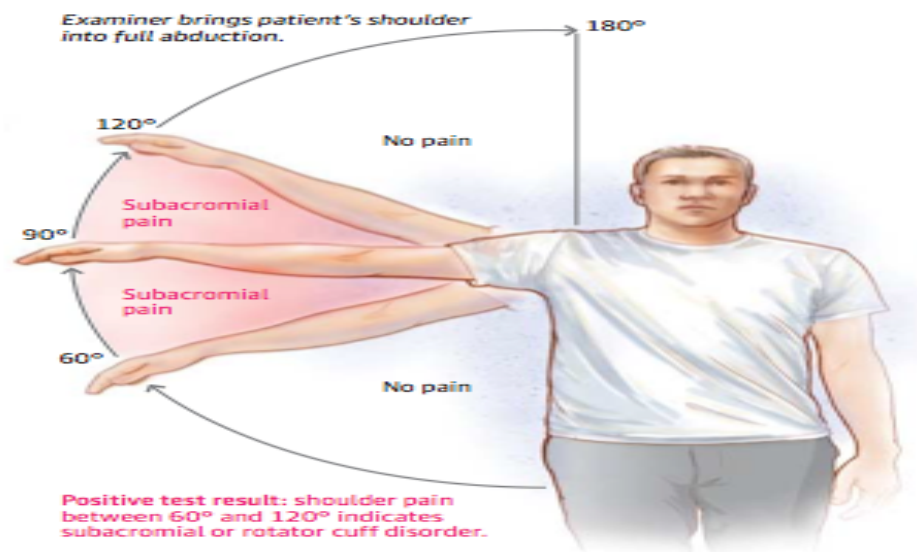
Front view means you can see both arms but depth is collapsed. If the arm drifts forward (flexion instead of abduction), it will appear to travel a shorter arc in the front view because you are losing the out-of-plane component. This is actually useful — you can detect forward drift by comparing expected arc length to actual arc length, but it also means MediaPipe elbow and wrist visibility may degrade if the arm moves too far forward out of the frontal plane.

Key Angles:

- 1) Shoulder Abduction Angle (Primary Tracking Angle)
 - Landmarks → hip (24/23) - shoulder (12/11) - elbow (14/13)
 - Start → ~160-175° (arm at side, minor variation per patient)
 - Peak → patient specific ~90° for standard abduction (arm horizontal) ~60-70° for restricted shoulder ~0-30° for full overhead abduction
 - Rep → peaks at resting arm-at-side position boundaries same logic as shoulder flexion

This is the tracking angle for rep detection. Peaks = arm at rest. Note: from a front view with the arm at the side, the hip-shoulder-elbow forms a nearly straight line (~170-175°). As the arm abducts outward, the elbow moves away from the body and this angle decreases toward 90° at

horizontal and further toward 0-30° at full overhead. The same direction of change as shoulder flexion — angle decreases as the arm rises.



2) Elbow Angle (Primary Correctness Feature)

- Landmarks → shoulder (12/11) - elbow (14/13) - wrist (16/15)
- Should stay → ~175-180° throughout the entire movement
- Same rule as shoulder flexion — the elbow must stay locked
- A bending elbow is always wrong regardless of height achieved

3) Trunk Lean (Compensation Angle)

- What it measures → does the patient tilt their trunk sideways away from the moving arm to gain extra range
- This is the abduction-specific compensation equivalent to the backward lean in shoulder flexion — different direction, same logic

Landmarks → left hip (23) - right hip (24) as a horizontal reference
 then track how much the shoulder on the moving side
 rises relative to the opposite shoulder

Two approaches:

Approach 1 — hip tilt angle (recommended)

Use both hip landmarks to measure how much the pelvis/trunk tilts

```
left_hip = np.array([r["left_hip_x"], r["left_hip_y"]])
```

```
right_hip = np.array([r["right_hip_x"], r["right_hip_y"]])
```

```
hip_vector = right_hip - left_hip
```

horizontal = np.array([1.0, 0.0])

hip_tilt = angle between hip_vector and horizontal

A tilting pelvis/trunk = lateral lean compensation

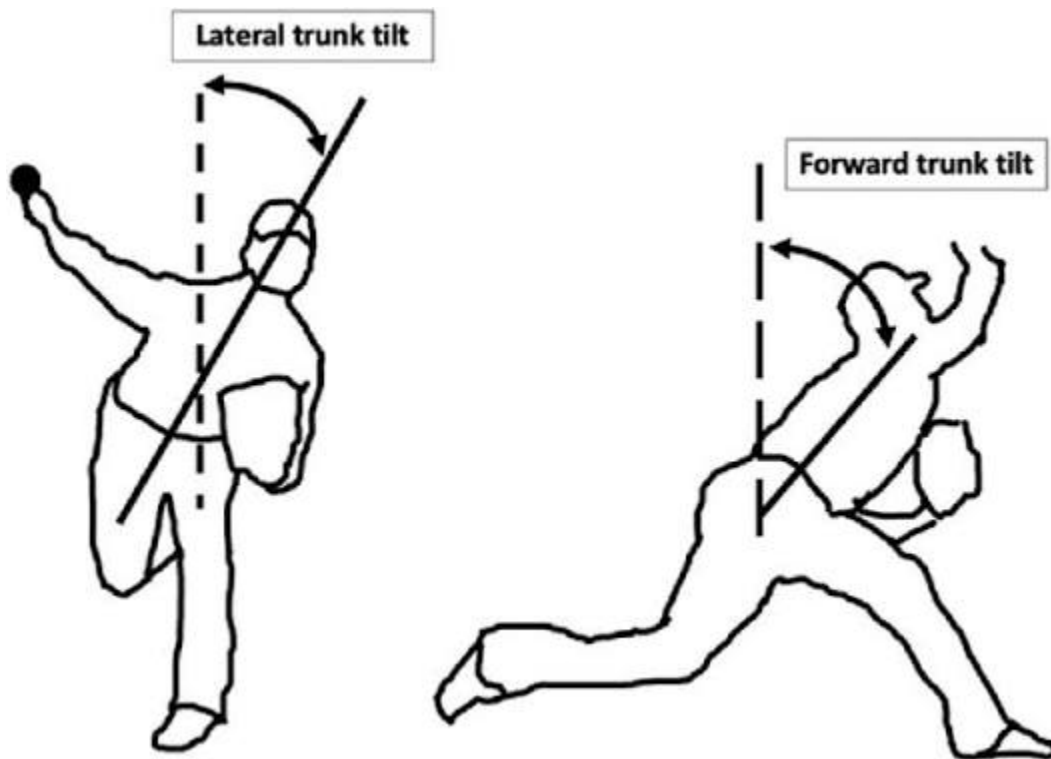
Approach 2 — bilateral shoulder height difference

Track the y-position of both shoulders each frame

shoulder_height_diff = left_shoulder_y - right_shoulder_y

If this value changes significantly during S2/S3 compared to S1 baseline, the trunk is tilting

Recommend Approach 1 — the hip landmark is more stable than the shoulder landmark during arm raise because the shoulder on the moving side itself rises with the arm, making bilateral shoulder comparison noisy during abduction.



4) Shoulder Shrugging (Secondary Correctness Feature)

- Same logic as shoulder flexion but even more important for abduction
- Abduction specifically loads the supraspinatus and middle deltoid — muscles that are among the most commonly weakened in shoulder pathology
- Shrugging recruits upper trapezius as a substitute

Track → vertical position (y) of the moving shoulder landmark

In S1 → record baseline shoulder_y

In S2/S3 → does shoulder_y decrease (rise toward ear) significantly relative to S1 baseline?

Normalise by torso length as in the flexion reference: $\text{shoulder_height} = (\text{hip_y} - \text{shoulder_y}) / \text{torso_length}$

5) Wrist Position Relative to the Elbow

- Landmarks → elbow (14/13) - wrist (16/15)
- From front view this is visible as whether the wrist drops below the elbow line during abduction
- Wrist dropping below elbow = elbow bending or wrist flexion Captured by elbow angle feature

States:

S1 → arm at side, shoulder angle ~170-175° (starting position)

S2 → arm rising outward, shoulder angle decreasing (ascending/abducting)

S3 → peak height, shoulder angle at minimum (held at top)

S4 → arm lowering inward, shoulder angle increasing (descending/adducting)

Common Errors:

1) Error 1 - Lateral trunk lean (most common, abduction-specific)

- What happens → patient tilts trunk sideways away from the raising arm — leaning to the left if the right arm is abducting, essentially tipping the body to create apparent extra range
- Why it happens → weak middle deltoid or supraspinatus cannot fully abduct the arm against gravity; patient compensates by using gravity and momentum from trunk tilt rather than true shoulder abduction
- Direction → trunk lean is LATERAL (sideways) — this is the key distinction from shoulder flexion where the compensation was BACKWARD lean
- Feature → trunk_tilt increases in S2/S3 relative to S1 baseline
S2_trunk_tilt_mean and S3_trunk_tilt_mean both elevated
- Universal? → yes, always wrong

2) Error 2 - Elbow bending

- What happens → elbow flexes partway through the raise
- Why it happens → shortens the lever arm to make the lift easier mechanically identical reason to shoulder flexion
- Feature → elbow_angle drops below 170° in S2/S3 S2_elbow_min, S3_elbow_min both reduced S2_elbow_std elevated (bending is gradual not sudden)
- Universal? → yes, always wrong

3) Error 3 - Shoulder shrugging

- What happens → shoulder elevates toward the ear as arm rises more common in abduction than flexion because supraspinatus weakness (extremely common in shoulder pathology) specifically causes compensatory trapezius recruitment during abduction
- Note → in abduction the first 15° of arm rise is driven almost entirely by supraspinatus (the "painful arc" of impingement is 60-120° of abduction specifically because this is where the supraspinatus tendon is maximally compressed).
- Shrugging is the body's workaround for supraspinatus pain or weakness.
- Feature → shoulder_height increases in S2/S3 relative to S1 (shoulder rising toward ear as arm rises)
- Universal? → yes, always wrong

4) Error 4 - Arm drifting forward (flexion instead of abduction)

- What happens → arm moves forward out of the frontal plane rather than staying directly to the side
- Why it happens → anterior deltoid (flexion) is stronger than middle deltoid (abduction) in many patients; the stronger muscle dominates the movement
Feature → from front view the arm appears to travel a shorter visible arc than expected — the elbow drifts slightly toward the camera which foreshortens the apparent abduction angle In practice: elbow_x displacement is less than expected for the achieved shoulder angle This is partially captured but imperfect from pure front view — confirm with your visualiser
- Universal? → yes, always wrong for abduction specifically
- Caveat → this is the hardest error to detect cleanly from a single front-view camera — include it as a class only after confirming it produces a detectable signal in your footage

5) Error 5 - Uncontrolled lowering

- What happens → patient drops the arm quickly on the way down
- Why does it happen → fatigue; or the patient does not understand that the eccentric lowering phase is equally important
- Feature → S4_shoulders_std elevated S4_duration significantly shorter than S2_duration $s2_s4_speed_ratio < 1$ (S4 much shorter)
- Universal? → yes, always wrong

Variables that might affect the shoulder abduction angles

1. Scapulohumeral Rhythm — The Most Important Abduction-Specific Variable What it is: Above approximately 90° of arm abduction, the scapula begins to rotate outward (upward rotation) in a fixed ratio to the glenohumeral motion. For every 2° of glenohumeral abduction above 90°, the scapula rotates roughly 1° — this is called the scapulohumeral rhythm, approximately 2:1. Below 90° most movement is glenohumeral; above 90° it is a combination.

Why it affects your angles: MediaPipe's shoulder landmark sits at the visible surface of the shoulder (the deltoid region). As the scapula rotates above 90° of abduction, the shoulder landmark physically moves — the whole shoulder complex rotates upward and outward. This means your hip-shoulder-elbow angle above 90° is not measuring purely glenohumeral abduction anymore; it is measuring a combination of glenohumeral and scapulothoracic motion. The angle still changes monotonically (keeps decreasing as arm rises), so rep detection is unaffected. But your absolute angle values in S3 for patients achieving overhead abduction reflect different anatomy than S3 for patients stopping at 90°.

How to handle it: Same principle as kyphosis — express features relative to S1 baseline. The scapulohumeral rhythm introduces a consistent, predictable bias in overhead patients; expressing features as change-from-baseline cancels most of it. Additionally: if your physio population mostly does not achieve overhead, this may not be a practical concern for your first model version.

2. The Painful Arc — A Clinical Feature That Is Also A Signal Issue

What it is: Between approximately 60° and 120° of abduction, the supraspinatus tendon passes under the acromion process. In patients with supraspinatus pathology (impingement, partial tear, tendinopathy — all extremely common in your patient population), this range is painful. Patients will often either speed through this zone (to get past the pain quickly) or show marked movement hesitation within it.

Why it matters for your angles: The painful arc will cause: Velocity irregularity in S2 and S4 specifically around 60-120° Possible elbow bending as the patient struggles through the arc Possible trunk lean beginning precisely within this arc

This means S2 and S4 angle standard deviations will be higher for impingement patients than for patients with clean motion — not because their form is technically wrong but because pain is disrupting smoothness. This is a clinically useful signal but you need to be aware that elevated std in S2/S4 may indicate pain rather than technique error, and your physio should be capturing pain scores alongside your video data.

How to handle it: Document whether each patient has known impingement as a metadata field. In your first model version, treat elevated S2/S4 std as an error signal regardless of cause — the model does not need to distinguish pain from poor technique for a classification task, it just needs to flag the rep as abnormal. The physio interprets why.